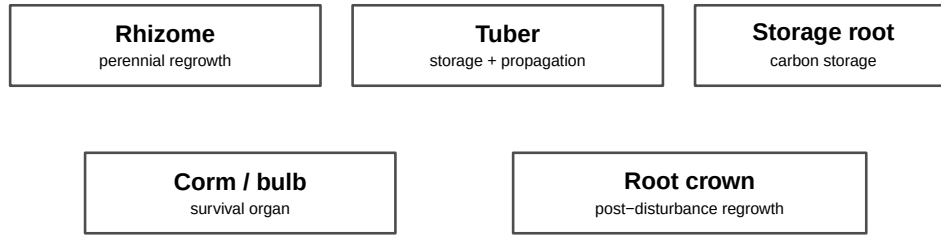


A

Biological scope

underground organs as survival, storage, and regrowth systems



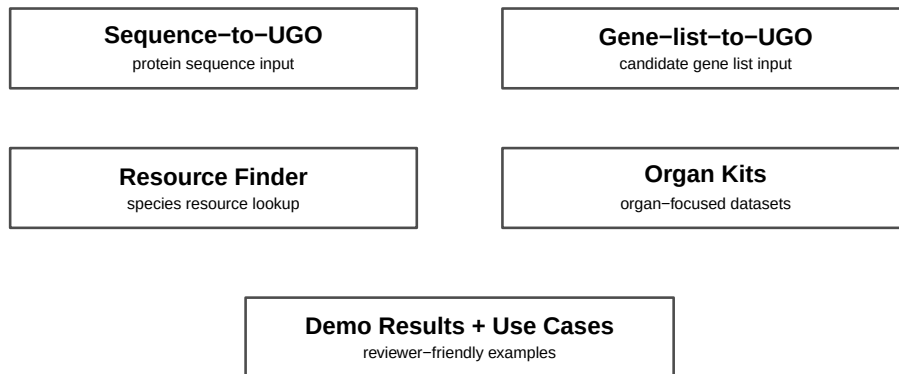
Shared demands:

persistence | regeneration | storage | developmental switching

C

Researcher-facing discovery tools

multiple entry points for underground-organ research

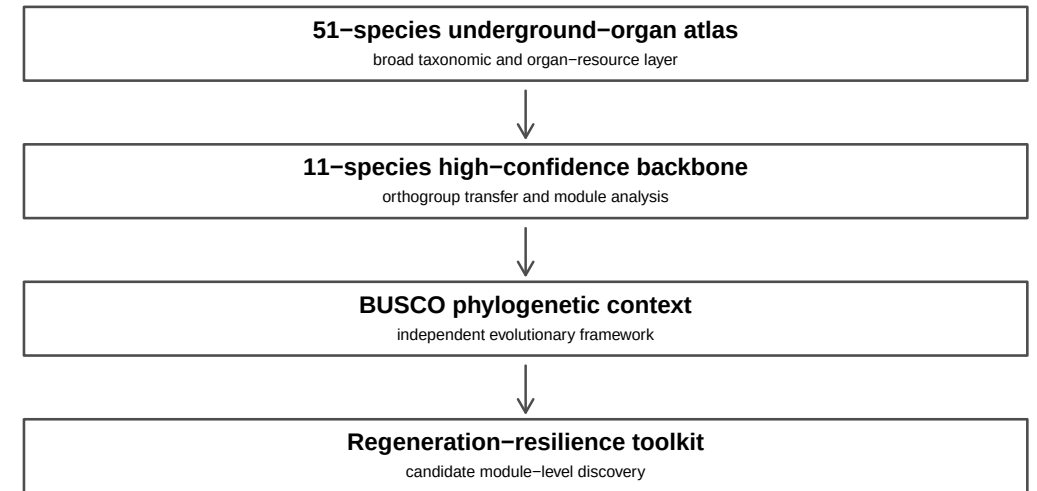


Output: candidate modules, orthogroups, downloadable resources, and testable hypotheses

B

Tiered data architecture

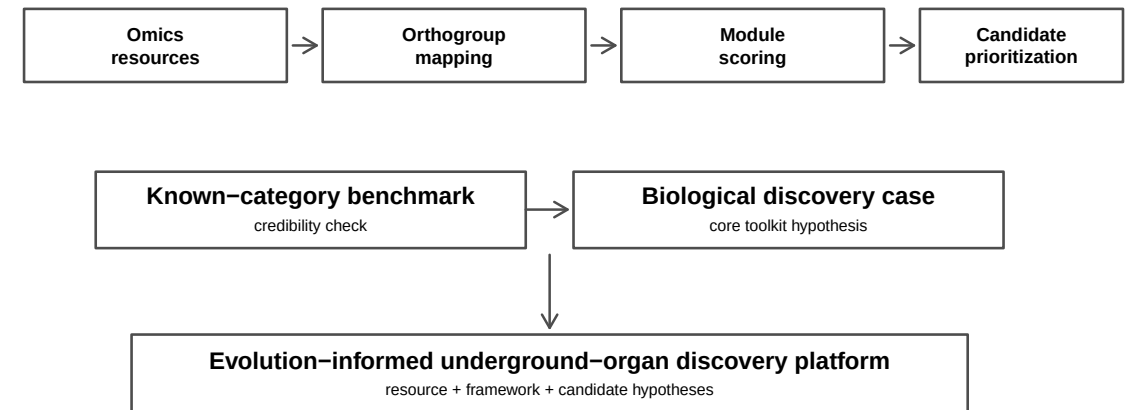
breadth, depth, and evolutionary context



D

UGO-Omics discovery logic

from data integration to biological hypotheses



Claim calibration: candidate prioritization and hypothesis generation, not direct functional validation.